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Application of a cellular automaton to simulations of granular flow in silos

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Abstract A cellular automaton based on a gas model of hydrodynamics was used to calculate the kinematics of non-cohesive granular materials during confined flow in a mass flow and funnel flow model silo. In the model, collisions of particles were taken into account during granular flow. In addition, a simplified automaton was used wherein granular flow was assumed as an upward propagation of holes through a lattice composed of cells representing single particles. The advantages and disadvantages of both cellular automata were outlined.

Keywords Cellular automaton · Collision · Density waves · Granular flow · Flow pattern · Lattice-gas automaton · Silo

1 Introduction

A cellular automata approach is a powerful method to describe, understand and simulate the behaviour of complex physical systems which are difficult to describe using traditional approaches (by means of differential equations). It is viewed as an alternative form of the microscopic reality which bears the expected macroscopic behaviour [1–2]. The concept dates back from the 1950s [3–4]. During the next fifty years of applications, cellular automata have been considerably improved (e.g. lattice-gas automaton, lattice Boltzmann method [1–2]) and used in many different fields of physics, chemistry and biology dealing with flow of fluid and gas [1–2], [5]. They have been also used to simulate the behaviour of granular materials during vibration, piling, toppling, segregation, displacement of a retaining wall and rapid flow [1], [6–25]. They introduced collision, friction, rotation and size of particles. They were used to simulate rapid flow in silos [8], [17–25] due to the fact that in this case, the silo fills above the outlet behave more as gas or fluid than as solid (although

the internal friction of the bulk material is still of a major importance [26]).

The cellular automata have both advantages and disadvantages relative to other modelling techniques. Among the advantages are the large number of particles, lack of restrictions for deformations, implementation simplicity and small amount of computer time needed to describe flow (the behaviour is described not by the terms of differential equations but by the terms of discrete systems). They provide insight into the main physical features of flow on both microscopic and macroscopic level. The main disadvantage is that the models are purely kinematics and no flow dynamics is involved.

The intention of simulations presented in this paper was to determine flow patterns of non-cohesive granular bodies in model silos. A realistic prognosis of the flow pattern of bulk solids in silos is very important when designing silos [27–31] since loads in silos are directly related to the flow type. Flow of stored bulk material in silos is of two main patterns: funnel flow (core flow) and mass flow. In mass flow, the entire material is in motion during discharge. In funnel flow, movement occurs only in a channel within the stored material, and this channel is surrounded by non-flowing material. Two different cellular models were applied using a two-dimensional array of hexagonal cells. In the first model based on a lattice gas automaton [8], [24], [32], inelastic collisions were taken into account. In the second simplified model presented in [15], [19–20], [25], granular flow was assumed as an upward propagation of holes through a lattice composed of cells representing single particles. The calculations were carried out with a mass flow and funnel flow model silo. The theoretical flow patterns were compared with experimental ones.

2 Description of the model

A cellular automaton requires: a) a lattice of adjacent cells covering a portion of a d -dimensional space, b) a set of variables attached to each site of the lattice and giving the local state of each cell at the time $t = 0, 1, 2, \dots$ and c) a rule which specifies the time evolution of the variables [1]. Thus,

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a cellular automaton is a system composed of adjacent cells (usually organised as a regular hexagonal, triangular or rectangular lattice) which evolves in a discrete time step. Each cell is characterised by an internal state whose value belongs to a finite set. The updating of these states is made in parallel according to local rules involving a neighbourhood of each cell.

Baxter et al. [16], [18] simulated a two-dimensional deterministic process of filling and flow of granular material in a hopper using a regular triangular array of cells. The direction of the displacement of grains corresponded to the direction in which the mechanical energy was minimal. The energy consisted of two terms: gravity and interaction between the neighbouring grains including friction, dilatancy and relative orientation of stick-like grains. Peng and Hermann [8] investigated density waves during granular flow in a vertical pipe with an improved cellular automaton. In their lattice-gas automaton model inelastic collisions of particles were taken into account explicitly conserving mass and momentum. During collisions, particles could change their velocities and during propagation move in the direction of their velocities to the neighbouring sites where they collided again. The four material parameters assumed which were related to gravity, dissipation, density and wall roughness. The model used by Martinez et al. [22–24] for silo flow was based on the transmission of the weight of grains with different migration rules (so called frontal, interstitial and constrictive).

Savage [19–20] used a simplified probabilistic cellular automaton with regular rectangular cells to describe two-dimensional flow in a hopper. An empty cell could be filled by particles from 3 cells located in a horizontal layer above. To model the internal friction angle of the silo fill, the aspect ratio of cells was changed. In the calculations, too large diffusion of particles was, however, observed. In a similar two-dimensional cellular model by Osinov [21] with a square grid of cells, the empty cell could be filled by particles from 3 cells (one was above, and two others at the sides). The transition probabilities, dimension of cells were the governing parameters affecting the hopper flow process. In addition, the simulations took into account a loosening of the material (the flowing particles could generate their copies). In a simplified cellular automaton model by Kozicki and Teichman [25], different migration schemes and transition probabilities were investigated during silo flow with an array of rectangular cells to realistically describe the angle of repose observed in the experiment. In addition, the effect of the outlet velocity and wall roughness on granular flow was simulated.

In all these simulations of silo flow [8], [17–25], the degree of agreement with laboratory tests was surprising in spite of the extreme simplicity of cellular automata models neglecting the dynamics of the granular flow process.

In the cellular automaton based on a lattice gas automaton [8], [32], some mechanical principles were taken into account. The model enables to investigate dilatant zones propagating inside of the silo fill during flow which are an inherent characteristics of each discharge process in mass flow and funnel flow hoppers and parallel-converging bins during

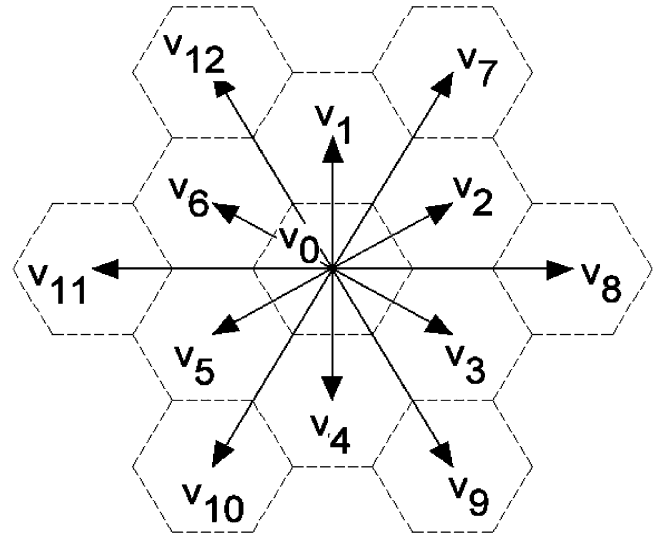


Fig. 1 Vectors of particle velocities at each site (lattice-gas automaton)

gravitational and controlled outflow of dense granular material composed of rough grains [16], [18], [33–38]. In this model, each site of a hexagonal lattice could be occupied by one or more particles, boundary wall or empty. The particles moved on the nearest-neighbour bonds (with unit velocity) and second-nearest neighbours of the lattice (with unit velocity multiplied by the factor $\sqrt{3}$). Each particle at each site had 13 Boolean states which were related to the velocity vectors v_i ($i = 0, 1, 2, \dots, 12$), Fig. 1. The direction of motion of a particle v could be toward any of its 12 nearest neighbours ($i = 1, 2, \dots, 12$) or be at rest ($v_0 = 0$). Thus, at the beginning of calculations, the number of particles per site had a maximum value of 13 and a minimal value of 0. The time evolution consisted of one collision step and two propagation steps. In the collision step, the particles both changed their velocities due to collisions or remained at rest in the site by losing the total kinetic energy due to dissipation. In the first propagation step, particles without a velocity (more particles than one) were scattered randomly to the nearest empty sites including low numbers of particles. In the second propagation step, colliding particles (after changing their velocities) were transferred in the direction of their velocities to the nearest sites where they collided again (Fig. 2). In

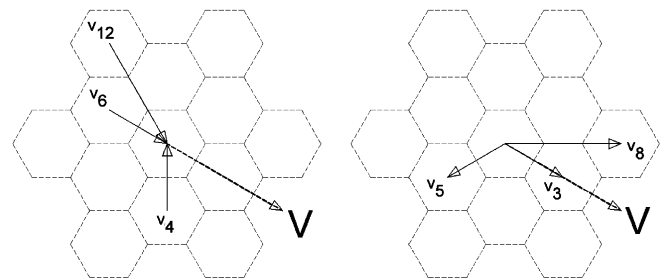


Fig. 2 Velocity vectors of particles after collisions using a lattice-gas automaton (V – resultant velocity)

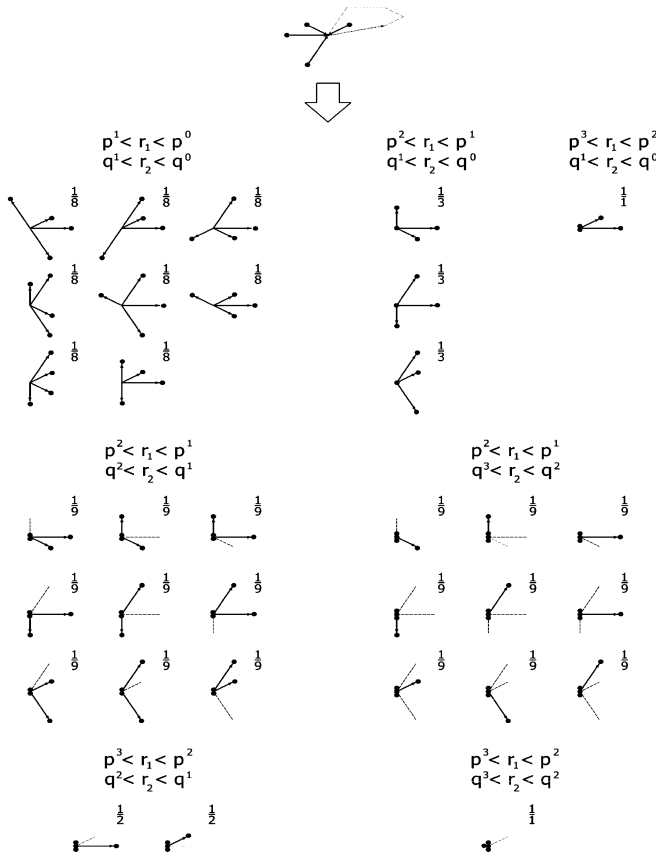


Fig. 3 Examples of collision rules for 4 particles flowing to one random site (arrows represent the directions of moving particles, dashed lines represent the unused directions due to dissipation and full dots stand for particles)

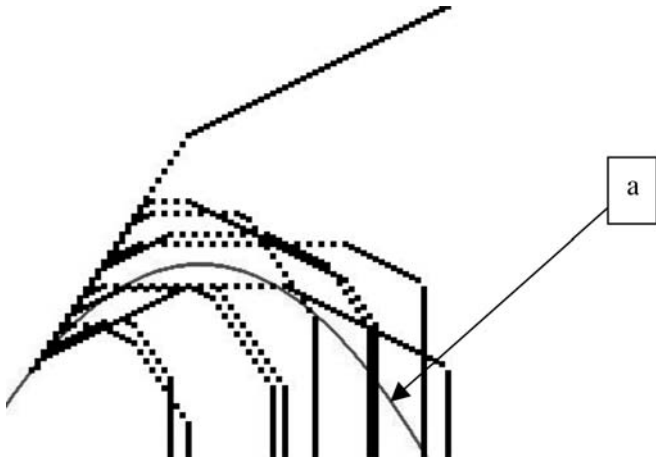


Fig. 4 Example traces of single particles in the gravity field thrown from the same point in the direction of v_7 of Fig. 1 (a – parabolic trajectory)

general, collisions conserved mass and momentum. The simulations showed that the calculation order of the lattice sites had no influence on results. The following six parameters were assumed in the model: two collision parameters (p, q), two friction parameters (b, k) and two gravity parameters

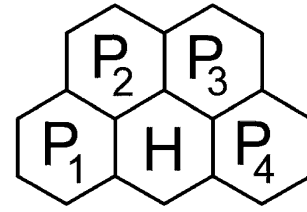


Fig. 5 Migration rule assumed for a two-dimensional hexagonal array in a simplified cellular automaton (H – void)

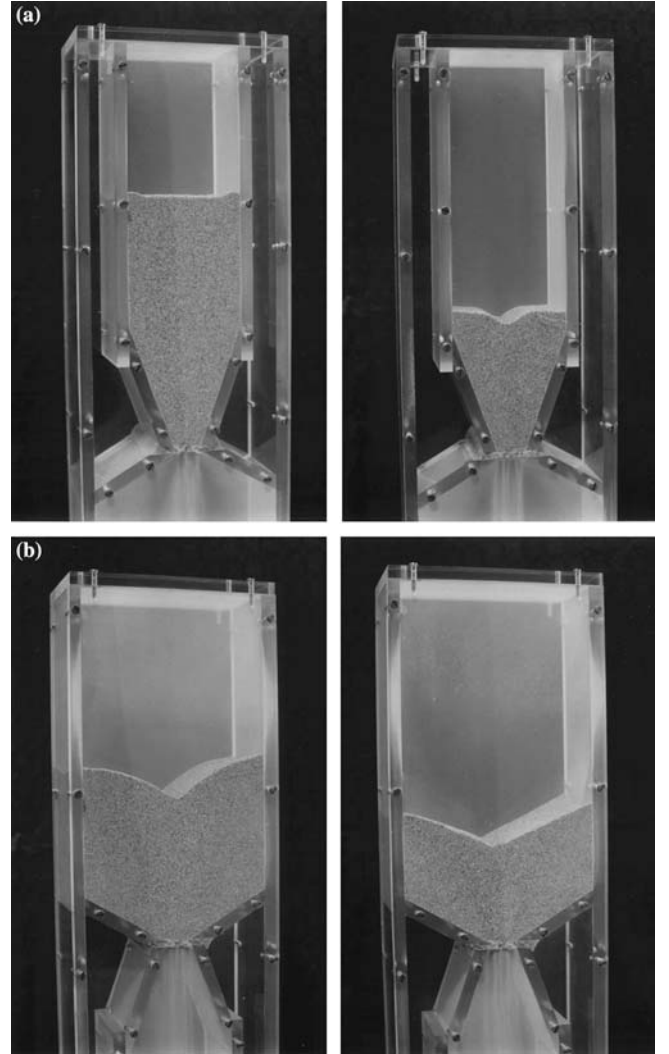


Fig. 6 Flow patterns observed in model silo tests: a) mass flow, b) funnel flow

(g, h). All probabilistic parameters were chosen in the range from 0 up to 1.

The collision parameter p determined the number of particles n which were at rest after the collision (due to the energy dissipation). The parameter n was calculated from the formula $p^{(n+1)} < r_1 < p^n$ (r is a random number). Due to the fact that some particles were stopped after collisions, other particles could leave the sites at a higher velocity due to the conservation of momentum. In the case, when

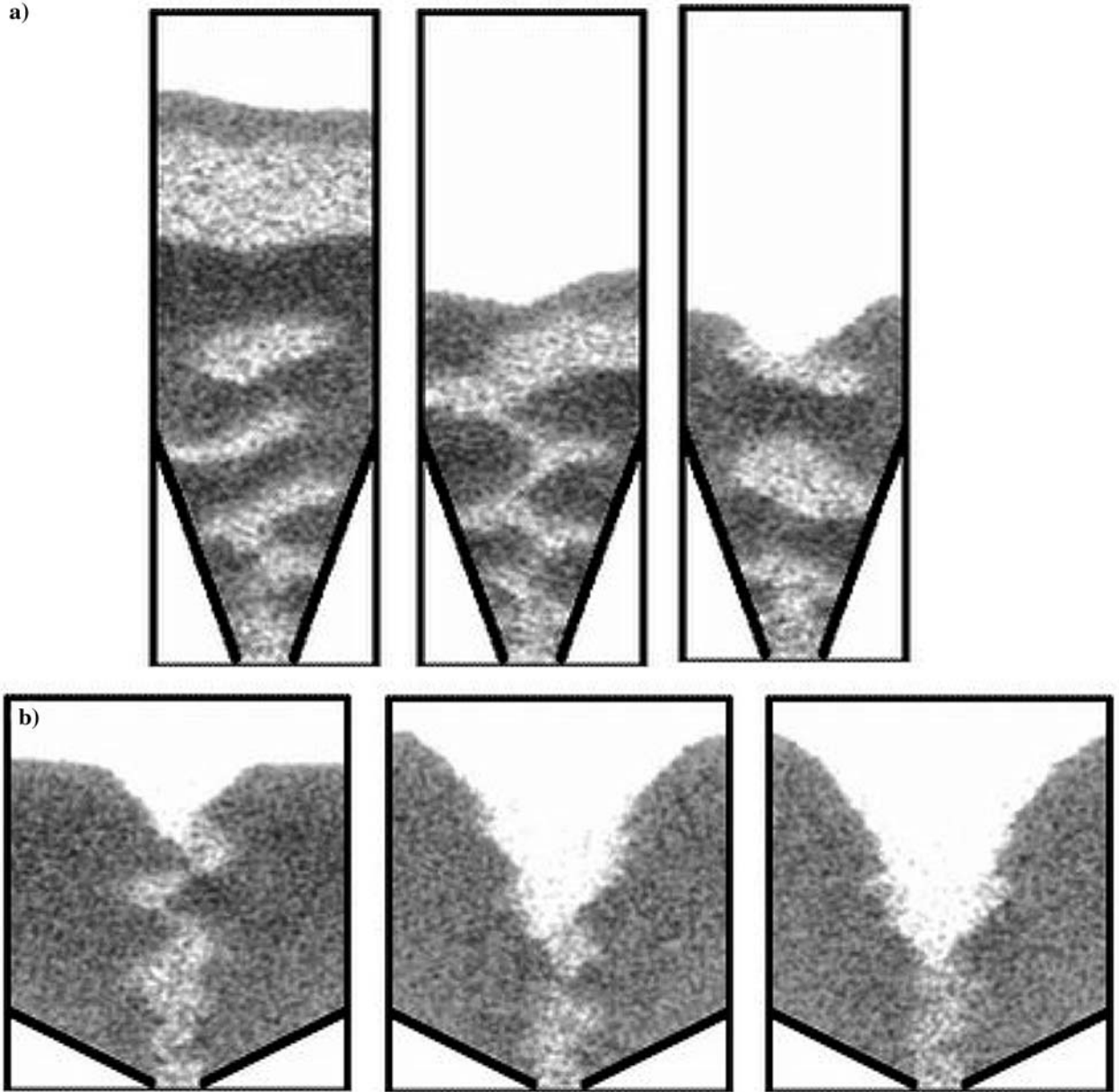


Fig. 7 Flow pattern in a mass (a) and funnel flow silo (b): $p = 0.2, q = 0.2, b = 0.05, k = 0.05, g = 0.2, h = 0.05$

more than one particle were stopped and the conservation of momentum could not be fulfilled, the particles moved in random directions. In turn, the parameter q described the loss of energy during each collision. The number of random particles l (which were not considered when calculating the resultant velocity) was obtained from the formula $q^{(l+1)} < r_2 < q^l$. Due to the loss of energy, the particles left the sites at a lower velocity. The total number of possible combinations for collisions was equal to $2^{12} = 4096$ and the number of possible velocity directions was 108. To speed up the calculation process, the appropriate table of collisions was generated (Fig. 3). If the parameters p and q were equal to zero, the sum of outgoing velocity vectors from each site was equal to

the sum of incoming velocity vectors (elastic collisions took place).

The friction parameter b described the wall roughness. It was introduced by two different methods. The first method was based on the model proposed in [8]. If the parameter b was different than zero (rough wall), the particles were bounced back from the wall at a random direction (with a probability of b). If the parameter $b = 0$ (smooth walls), the angle at which the particles hit the wall was equal to the angle at which they were reflected. In the second method, the particles hitting the wall were stopped there with a probability of b (the increase of wall roughness corresponded to the increase of the parameter b). Afterwards, they were

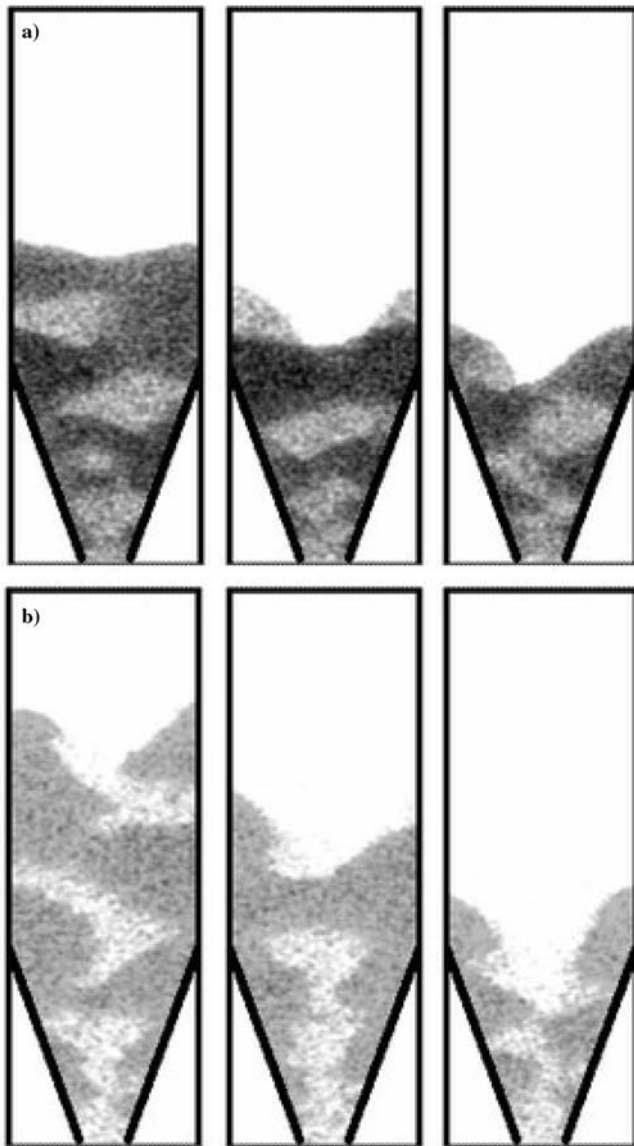


Fig. 8 Flow pattern in a mass flow silo (effect of the gravity parameter g): a) $p = 0.2, q = 0.2, b = 0.05, k = 0.05, g = 0.95, h = 0.05$, b) $p = 0.2, q = 0.2, b = 0.05, k = 0.05, g = 0.05, h = 0.05$

scattered randomly to the nearest empty sites with a low contents of particles. The friction parameter k described the internal interaction between particles. When the parameter k was different than zero, the certain migration directions became dominant (to induce movement in one specific direction). In our simulations, the horizontal direction was assumed to be dominant. Thus, the resulting velocity vector of outgoing particles was replaced by the sum of random velocity vectors in the direction v_2, v_3, v_5, v_6, v_8 and v_{11} with a probability of k (Fig. 1).

The gravity parameter g was taken into account during each collision in this way that the vertical velocity v_4 (Fig. 1) was added to the resulting velocity vector of outgoing particles with a probability of g . Thus, the particles were

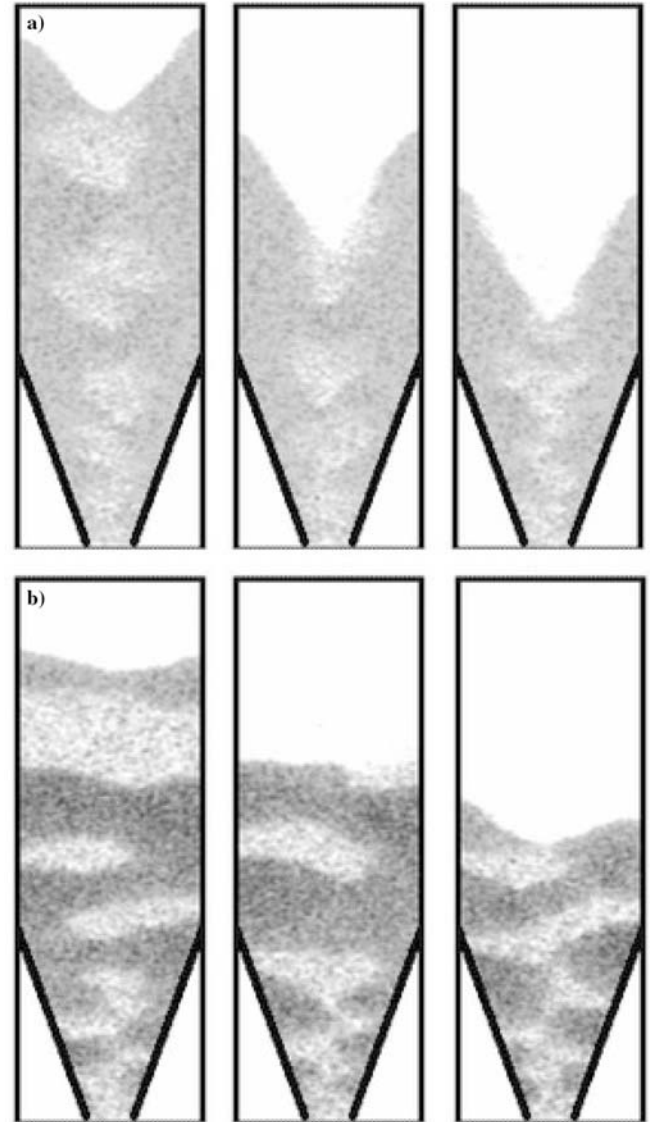


Fig. 9 Flow pattern in a mass flow silo (effect of the gravity parameter h): a) $p = 0.2, q = 0.2, b = 0.05, k = 0.05, g = 0.2, h = 0.8$, b) $p = 0.2, q = 0.2, b = 0.05, k = 0.05, g = 0.2, h = 0.0$

accelerated downwards after each collision. In turn, the gravity parameter h was introduced to improve the direction of the resultant velocity vector so that the direction would become parabolic for single particles (Fig. 4). The particle at rest at the given site (v_0) could move downwards (with the probability of h) to the state v_0 in the neighboring empty site without any velocity only in one of the chosen directions v_3, v_4, v_5, v_9 and v_{10} (Fig. 1). If $h = 1$, the particles without velocities moved downwards (density became more uniform). In turn, if $h = 0$, the particles without velocities did not move.

The initial density of the granulate at the onset of flow depending upon the parameters p, q, g and k was generated during silo filling when the outlet was closed. The granular material after a settlement in the silo reached a certain density

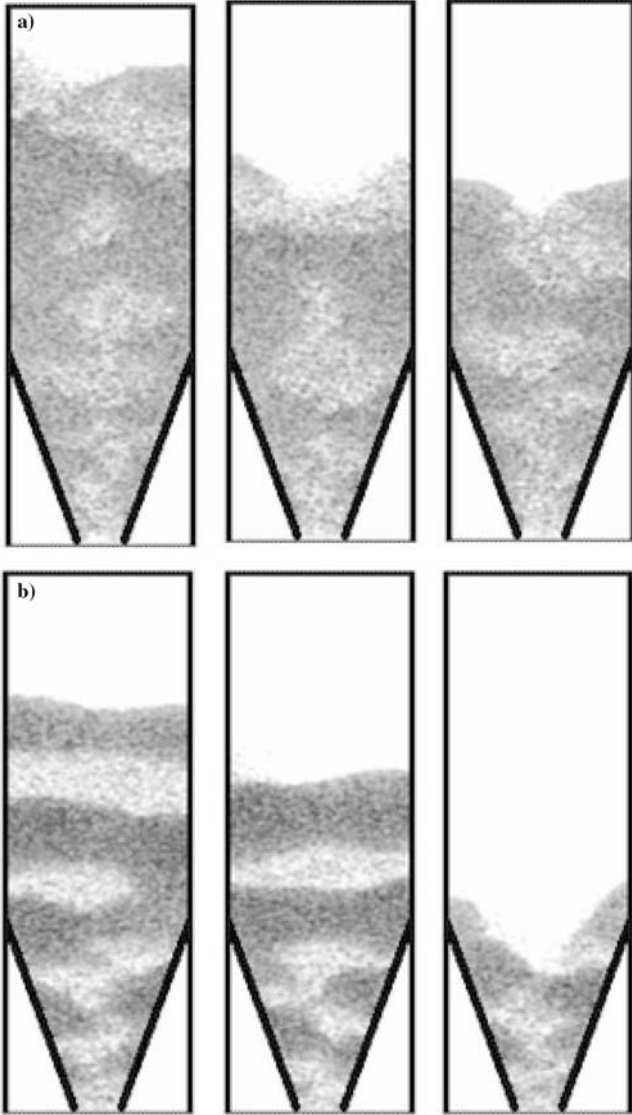


Fig. 10 Flow pattern in a mass flow silo (effect of the friction parameter k): a) $p = 0.2, q = 0.2, b = 0.05, k = 0.8, g = 0.2, h = 0.05$, b) $p = 0.2, q = 0.2, b = 0.05, k = 0.0, g = 0.2, h = 0.05$

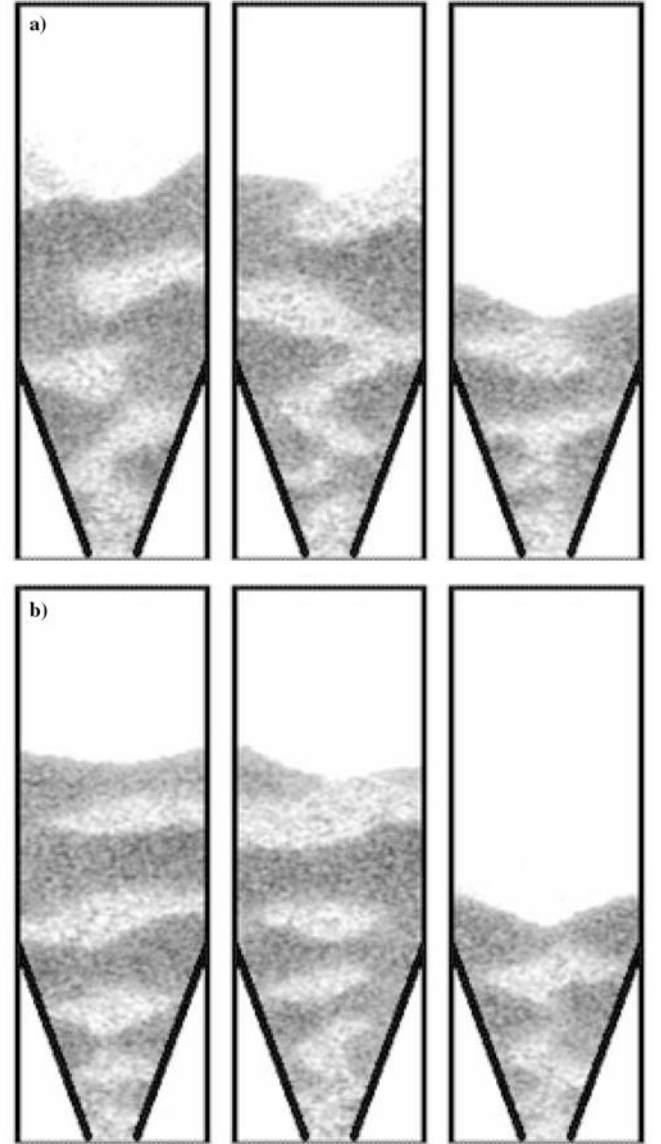


Fig. 11 Flow pattern in a mass flow silo (effect of the wall roughness parameter b): a) $p = 0.2, q = 0.2, b = 0.8, k = 0.05, g = 0.2, h = 0.05$, b) $p = 0.2, q = 0.2, b = 0.0, k = 0.05, g = 0.2, h = 0.05$

which increased with increasing p and g and decreasing q and k .

The model used is based on the lattice-gas automaton proposed for granular materials in [8] and [32]. The novelties were following: the model was applied to silo flow, the number of possible vector directions was enhanced (13 instead of 7), the material density was not used as an additional parameter, gravitation was depicted with two parameters, rough walls were able to stop flowing particles, the method to describe collisions allowed to stop any number of particles, and the new parameter was introduced to describe the internal interaction between particles.

In the simplified cellular automaton applied, granular flow of separate particles in silos was described as an upward propagation and diffusion of holes through a lattice of

cells (particles moved downwards and voids moved upwards) [19–21], [25]. Each cell might be empty or filled by only one particle of the material. The computation process involved searching through all lattice cells for holes. At each time step, a particle flowing downwards due to gravity could move from one cell to an empty neighbouring cell or stay at rest. The filling of one cell caused the formation of a new empty one, which was again filled with a randomly chosen particle. Thus, the migration of holes inside the granular materials occurred. The transition of holes corresponded to the transport of the material in the opposite direction. The evolution rule describing the state change was probabilistic since in a real bulk solid the shape and dimensions of particles, contact points and forces between particles, and particle roughness are completely random. Fig. 5 shows a migration rule using

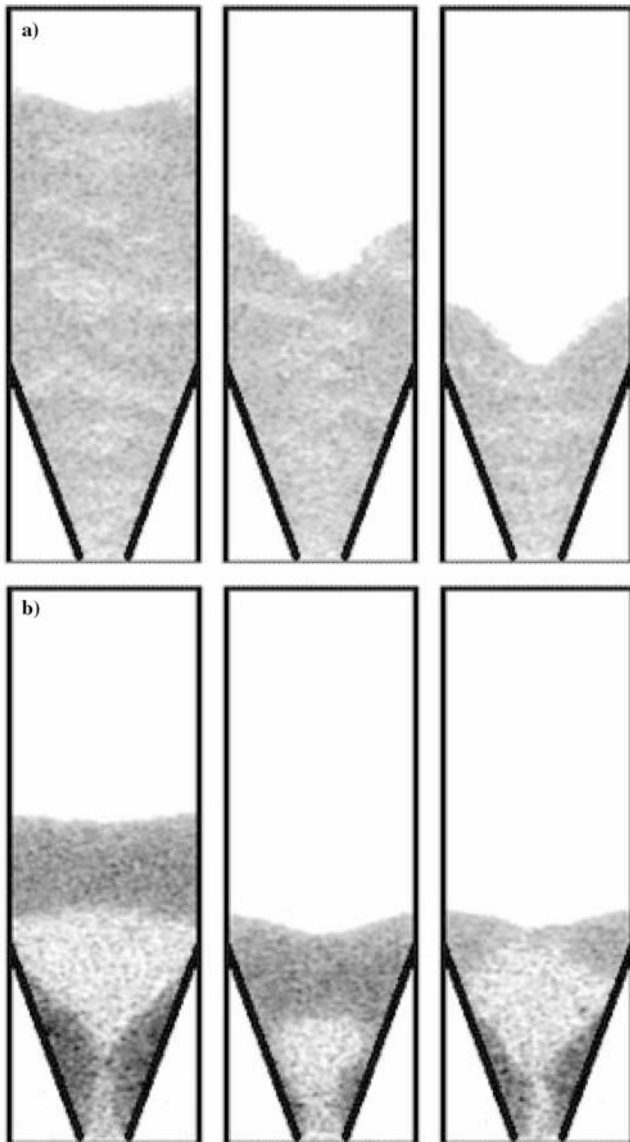


Fig. 12 Flow pattern in a mass flow silo (effect of the collision parameter p): a) $p = 0.8$, $q = 0.2$, $b = 0.05$, $k = 0.05$, $g = 0.2$, $h = 0.05$, b) $p = 0.0$, $q = 0.2$, $b = 0.05$, $k = 0.05$, $g = 0.2$, $h = 0.05$

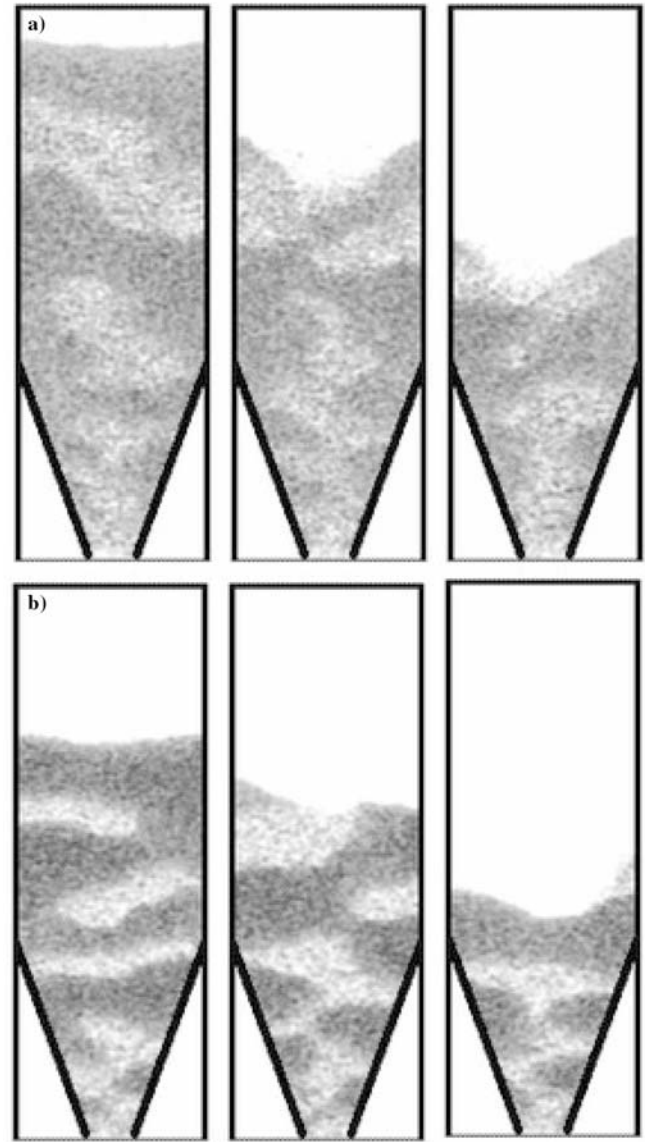


Fig. 13 Flow pattern in a mass flow silo (effect of the collision parameter q): a) $p = 0.2$, $q = 0.8$, $b = 0.05$, $k = 0.05$, $g = 0.2$, $h = 0.05$, b) $p = 0.2$, $q = 0.0$, $b = 0.05$, $k = 0.05$, $g = 0.2$, $h = 0.05$

a hexagonal lattice. A hexagonal grid is a better choice to simulate a Navier-Stokes equation of fluid flow because of isotropy of the momentum advection tensor [1]. An empty cell could be filled by particles from two cells located above it and two at sides. The probability of the hole transport from one cell to another was assumed to be p_i ($\sum p_i = 1$) and did not depend on coordinates. In this model, the propagation of density waves could not be simulated.

3 Model experiments

The simple laboratory model tests carried out with a mass flow silo and a funnel flow silo made of perspex. The dimensions of a mass flow silo were: 0.32 m (height), 0.09 m (width)

and 0.07 m (length) and of a funnel flow silo: 0.28 m (height), 0.15 m (width) and 0.07 m (length). The wall inclination of the hopper against the bottom was 60° (mass flow silo) and 30° (funnel flow silo). The silos contained dry sand with a mean grain diameter 0.5 mm. The outflow was due to gravitation. The experimental flow patterns during both mass and funnel flow are demonstrated in Fig. 6.

4 Simulation results

4.1 Lattice-gas automaton

The 2D-results of simulations with a lattice-gas automaton including 35000 cells are demonstrated in Figs. 7 (mass flow

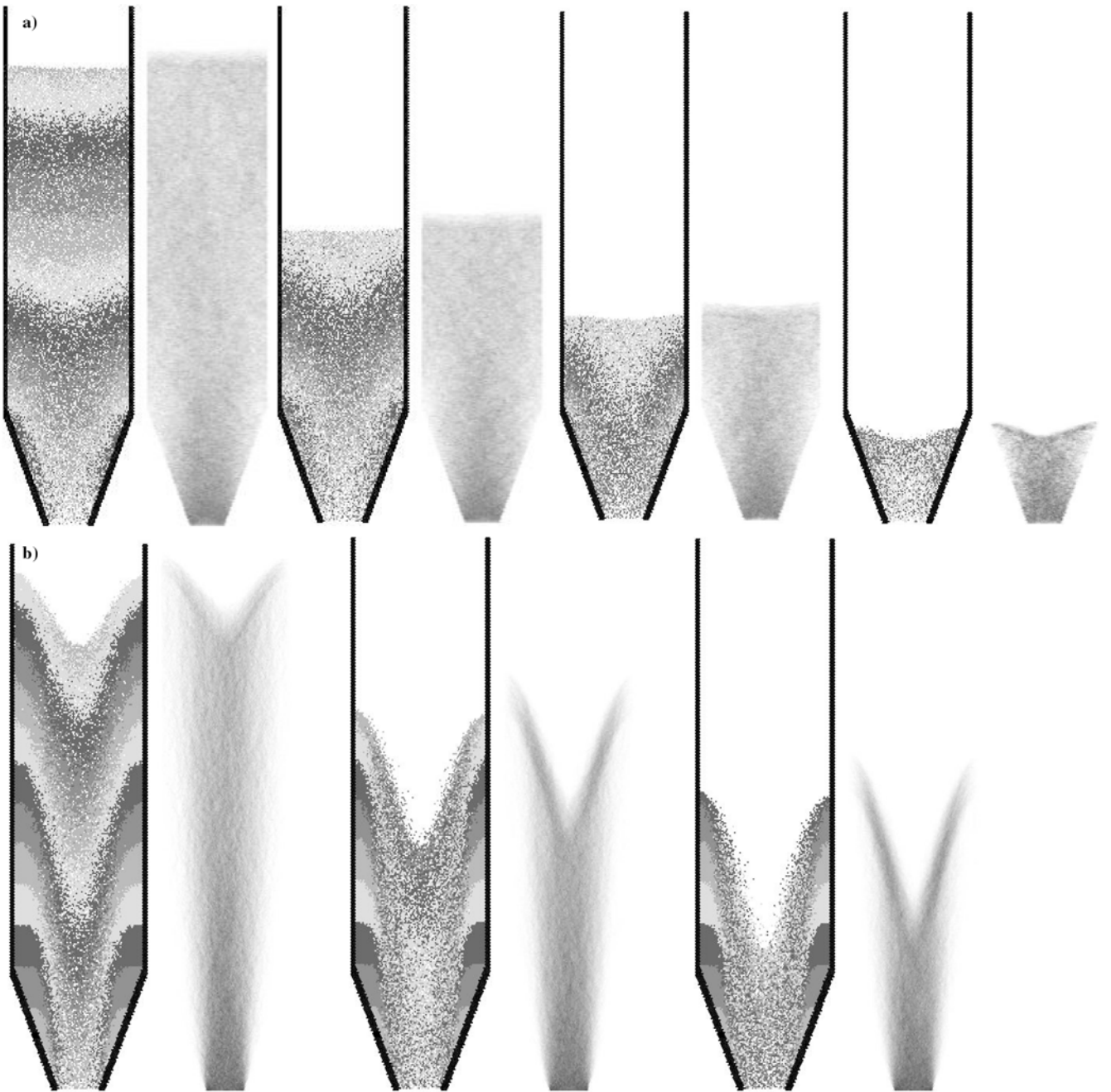


Fig. 14 Flow patterns and distribution of flow rate during granular flow in a mass flow silo: a) $p_1 = p_4 = 0.4$, $p_2 = p_3 = 0.1$, b) $p_1 = p_4 = 0.1$, $p_2 = p_3 = 0.4$ (simplified cellular automaton)

silos of Fig. 5a) and 8 (funnel flow silo of Fig. 6). The values of the parameters were assumed to be random: $p = 0.2$, $q = 0.2$, $b = 0.05$, $k = 0.05$, $g = 0.2$ and $h = 0.05$. The darker regions are associated with the higher densities, and the lighter regions are associated with the lower densities of the silo fill. In the calculations with the parameter b describing the wall roughness, the second method was applied (the first one, in contrast to results in a vertical pipe [8], did not affect the results in silos).

The effect of the parameters g , h , k , b , p and q is shown only for a mass flow silo (Figs. 8–13); their effect in a fun-

nel flow silo was similar. For a better comparison, one chose small and large values of these parameters.

In all cases, regions with a different density occur during granular flow. The shape of dilatant zones is similar for both flow types and is strongly influenced by the collision parameter p and gravity parameter h . In a mass flow silo, it is also dependent upon the wall roughness parameter b (more particles were stopped at the wall). With decreasing collision parameter p , the height of dilatant zones significantly increases (material behaves more as gas). If the gravity parameter h is large, the dilatant zones become narrower and are

more concentrated in the mid-region of the silo due to the fact that there are more empty sites in the middle of the silo than close to walls.

For a smaller wall roughness parameter b , the dilatant zones are more horizontal since the flow disturbance close to walls is smaller. The outflow rate increases obviously with increasing gravity parameter g . With an increase of the gravity parameter g , the effect of other factors becomes also insignificant. The inclination of the upper free surface of the material decreases with decreasing parameters p , q and h (material behaves more as gas). The non-uniformity of the material density increases with increasing parameter g and decreasing parameters h , k , p and q .

The calculated flow patterns in silos (Fig. 7) are in a satisfactory agreement with the experimental ones of Fig. 5. However, the shape of the propagating dilatant zones inside of the flowing material differs from the experimental ones due to the fact that pure shearing of the granular material was neglected. In the experiment with a mass flow silo [36–37], a symmetrical pair of curvilinear dilatant rupture zones was created in the neighbourhood of the outlet. The zones propagated upward, crossed each other around the symmetry of the silo, reached the walls and subsequently were reflected from them. This process repeated itself until the zones reached the free boundary in the converging hopper or the transition zone in the parallel-converging silo. In the case of funnel flow silo [16], [18], the curvilinear dilatant zones in the material core were symmetric about a vertical mid-line. Some of them crossed each other.

4.2 Simplified cellular automaton

The 2D-results with a simplified cellular automaton (using 50000 hexagonal cells) are shown in Figs. 14–16. To visualize the calculation process, horizontal layers of a various shade are introduced. The calculations were carried out with a migration rule of Fig.5, and two different distributions of transition probability values p_i ; larger ones in the cells at sides of the void and smaller ones in the cells above the void (Figs.14a and 15a) or inversely (Figs.14b and 15b). The probability values were assumed to be random: a) $p_1 = p_4 = 0.4$ and $p_2 = p_3 = 0.1$, and b) $p_1 = p_4 = 0.1$ and $p_2 = p_3 = 0.4$. In addition, the distribution of the flow rate was shown, expressed by the overall number of grains going through the single cells (the darker the region, the greater the flow rate).

The agreement between the calculated flow pattern (Figs.14a and 15a) and experimental one (Fig.6) is satisfactory. To obtain a better agreement, the transition probabilities should be more carefully fitted to the experimental results. The angle of repose (inclination of the upper free boundary) becomes smaller when the transition probabilities are the largest at sides and becomes larger when the transition probabilities are the smallest at sides. The flow rate evolution at the outlet during silo discharge is presented in Fig.16. The mean flow rate was described by the overall number of cells coming out from the silo outlet per iteration. At the beginning of discharge, a constant flow rate was observed during

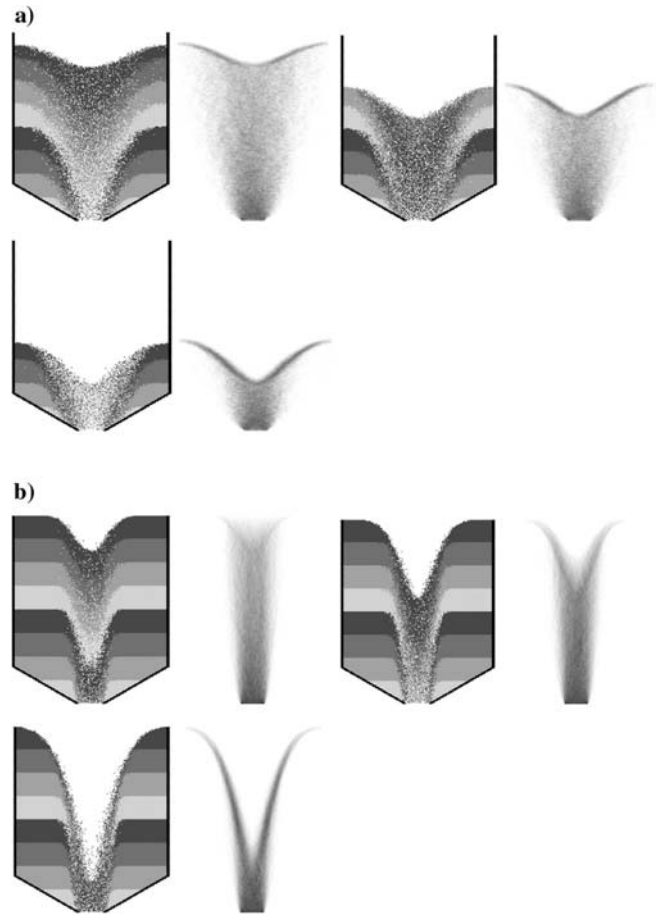


Fig. 15 Flow patterns and distribution of flow rate during granular flow in a funnel flow silo: a) $p_1 = p_4 = 0.4$, $p_2 = p_3 = 0.1$, b) $p_1 = p_4 = 0.1$, $p_2 = p_3 = 0.4$ (simplified cellular automaton)

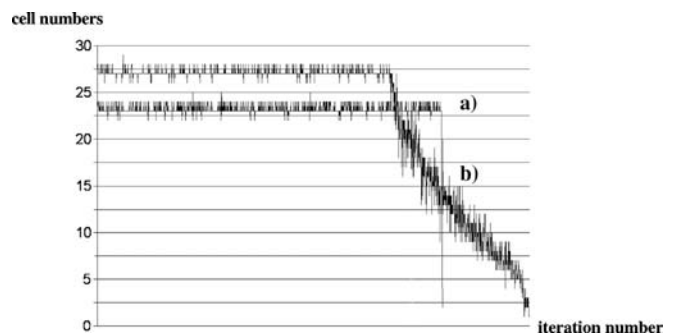


Fig. 16 Flow rate evolution: a) mass flow silo, b) funnel flow silo (simplified cellular automaton)

both types of flow what is in accordance with physical experiments [30–31]. In the end phase of funnel flow, a non-linear reduction of the flow rate was obtained due to a successive decrease of the material contents in the silo. In the case of mass flow, a drastic reduction of the flow rate took place. The calculation time for silo flow with 50000 cells using a PC 2.0 GHz was about 2 minutes.

5 Conclusions

Cellular automata are purely kinematic models wherein flow dynamics is not taken into account. Although they are a gross oversimplification of a real granular material, they were apt to realistically describe the flow pattern of granulates in silos on the basis of a back analysis of laboratory experiments.

The lattice-gas automaton was able to describe the dilatant zones during silo flow. Their shape was affected by gravity, collision and friction parameters. However, it was different than this obtained in the experiments.

The main parameters governing the motion of granular particles in silos in the simplified cellular automaton were transition probabilities. A decrease of transition probability values from the cells at sides of the void towards the cells above the void was able to capture the angle of repose of the bulk solid. This automaton can be used to approximately and quickly determine the flow patterns of non-cohesive bulk materials in silos in the phase of the silo design.

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